Day 87: Embedding Visualization — PCA and t-SNE From Scratch, and What Survives the Projection

MD Mutasim Billah | 52-Week Data Science to ML/AI Roadmap | Week 13, Day 4 | Complete study guide

Watch: Principal Component Analysis (PCA), Step-by-Step (StatQuest) | t-SNE, Clearly Explained (StatQuest)

Script: `day87_embedding_visualization.py` · every number in this guide is copied by program from `day87_run_output.txt`, the real output of that script.

Day 86's word vectors have 50 numbers each. Two ways to draw them in two dimensions, both written from scratch and each checked by a number, not by eye. PCA keeps the directions of largest spread, but loses the word groups: only 0.433 of category words keep a same-group nearest neighbour, against 0.967 in 50-D. t-SNE keeps them (0.944 to 0.956 over five random starts). Two controls show what t-SNE pictures cannot be trusted with: the sizes of groups and the distances between them.

0. Where today fits

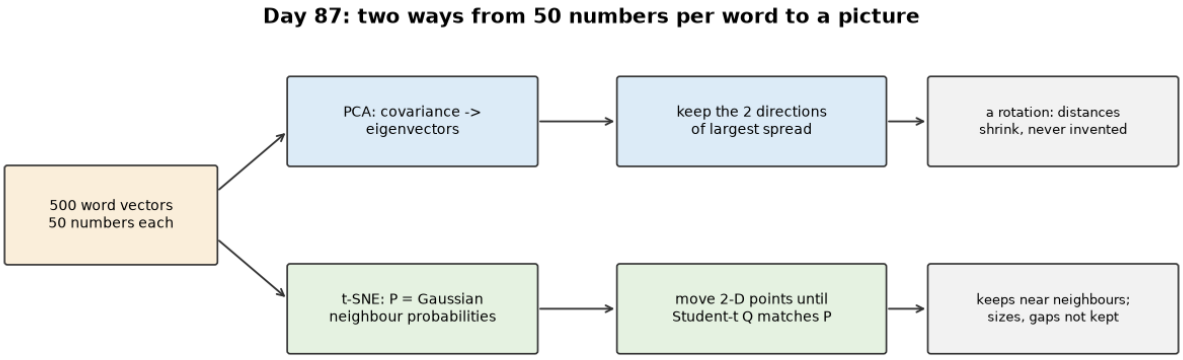
DEFINITION — start here, no prior knowledge assumed

A **projection** turns many numbers per point into two, so the points can be drawn. **PCA** (principal component analysis) finds the directions along which the points spread the most and keeps the first two: a rotation of the data followed by dropping the other axes. **t-SNE** (t-distributed stochastic neighbour embedding) places points in 2-D so that each point's near neighbours in the original space stay near; it does not try to keep large distances.

REAL-WORLD SCENARIO

A world map. A flat map cannot keep every distance of a globe: some keep shapes, some areas, none all of it. Which map is right depends on the question.

SYSTEM DIAGRAM



PCA and t-SNE, side by side.

Today's part for Day 90

`day87_exercise.py` is `pca` in one function, checked against NumPy's SVD. Day 90 uses it to draw its learned character table.

How to read each section

Step	What it gives you
DEFINITION	What the idea is, in plain words. Start here.
REAL-WORLD SCENARIO	An everyday situation where the same idea shows up.
MATHEMATICS	The formula, when the section introduces one.
SPECIFICATION	The exact setup of the experiment.
Why this test	The question the experiment answers.
IMPLEMENTATION	The real code from the script.
ACTUAL OUTPUT	What the script really printed, copied unchanged.
What it means	How to read the numbers, and what they do and do not show.
Why choose this? / Why NOT this?	When the idea is the right tool, and what it costs.

1. PCA from scratch

DEFINITION — start here, no prior knowledge assumed

Centre the data, compute the covariance matrix (how every pair of columns varies together), and take its eigenvectors: the directions of largest spread, largest first. Each eigenvalue is the spread along its direction.

REAL-WORLD SCENARIO

Photographing a long, thin object: you turn it until the camera sees its longest side.

MATHEMATICS

$$C = X_c^T X_c / (n-1); \quad C v_i = \lambda_i v_i; \quad Z = X_c [v_1 v_2]; \quad \text{share kept} = (\lambda_1 + \lambda_2) / \Sigma \lambda$$

IMPLEMENTATION

```
def pca(X, k):
    Xc = X - X.mean(axis=0)
    C = Xc.T @ Xc / (len(X) - 1)
    vals, vecs = np.linalg.eigh(C)
    order = np.argsort(vals)[::-1]
    vals, vecs = vals[order], vecs[:, order]
    return Xc @ vecs[:, :k], vecs[:, :k], vals / vals.sum()
```

ACTUAL OUTPUT

```
21681 words x 50 numbers (Day 86's table, rows scaled to length 1)
largest |difference| from SVD: directions 9.9e-14, projections 7.1e-14
share of the spread kept: 2 directions 0.377, 10 directions 0.823, 25 directions 0.959
the largest direction holds 0.240; a direction of 50 equally spread ones would hold 0.020
Actual output from this lesson's run.
```

What it means

The result matches NumPy's SVD to rounding (9.9e-14). Two directions keep 0.377 of the spread and ten keep 0.823: the vectors are not flat. The largest direction holds 0.240, far more than the 0.020 an evenly spread direction would hold, so the table does have a few dominant directions.

WHY CHOOSE THIS?

Exact, fast, repeatable, and distances only shrink: nothing in the picture is invented.

WHY NOT THIS? (TRADEOFFS)

Two directions here keep under 40% of the spread; what is lost may be the structure you care about.

2. t-SNE from scratch

DEFINITION — start here, no prior knowledge assumed

t-SNE turns distances into neighbour probabilities, p_{ij}, with a Gaussian around each point whose width is set so that each point has about 'perplexity' effective neighbours. In 2-D it uses a heavier-tailed Student-t curve, q_{ij}, and moves the points downhill on the mismatch KL(P || Q).

REAL-WORLD SCENARIO

Seating guests so that friends sit together; how far apart two unrelated tables are does not matter.

MATHEMATICS

$$p_{j|i} \propto exp(-\beta_i |x_i - x_j|^2),\ \beta_i \text{ set by perplexity};\quad p_{ij} = (p_{j|i} + p_{i|j}) / 2n$$

$$q_{ij} \propto (1 + |y_i - y_j|^2)^{-1};\quad KL = \Sigma\, p_{ij}\, log(p_{ij} / q_{ij});\quad dKL/dy_i = 4\, \Sigma_j\, (p_{ij} - q_{ij})(1 + |y_i - y_j|^2)^{-1}(y_i - y_j)$$

IMPLEMENTATION

```
def kl_and_grad(P, Y):
    num = 1 / (1 + sq_dists(Y)); np.fill_diagonal(num, 0)
    Q = np.maximum(num / num.sum(), 1e-12)
    kl = np.sum(P * np.log(np.maximum(P, 1e-12) / Q))
    W = (P - Q) * num
    grad = 4 * (W.sum(axis=1)[:, None] * Y - W @ Y)
    return kl, grad
```

ACTUAL OUTPUT

```
gradient check: 16 coordinates, largest |numeric - analytic| = 1.24e-10, all pass: True
P sums to 1.000000; symmetric: True
Actual output from this lesson's run.
```

What it means

The gradient matches the numerical one (1.24e-10), and P is a proper symmetric probability table. The optimiser itself (early exaggeration, momentum, per-coordinate gains) follows van der Maaten's reference code.

WHY CHOOSE THIS?

A checked loss: the pictures come from the method as published.

WHY NOT THIS? (TRADEOFFS)

Exact t-SNE costs n^2 per step; 500 points is fine, 50,000 is not.

3. 500 words in two dimensions

DEFINITION — start here, no prior knowledge assumed

Day 86's 90 category words plus the 410 most frequent other words. Two numbers per picture: the **group check** (does each category word's nearest category word share its group?) and **preserved** (of each word's 10 nearest neighbours in 50-D, how many are still among its 10 nearest in 2-D?).

REAL-WORLD SCENARIO

Judging a map by measuring it: how many neighbouring towns are still neighbours on paper?

SPECIFICATION

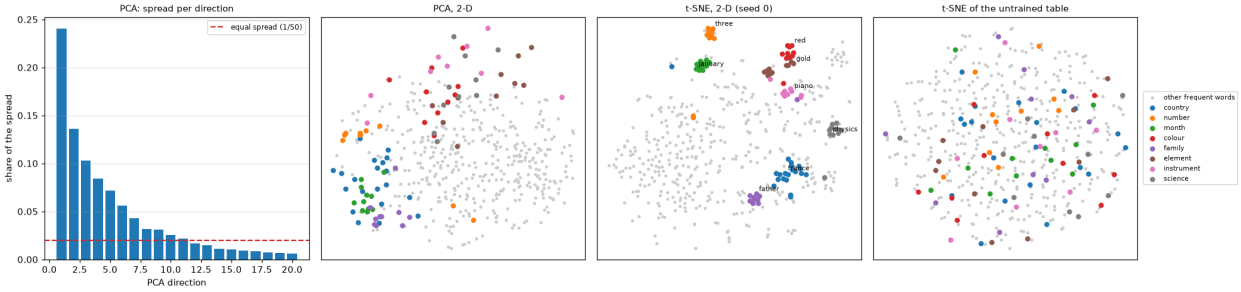
Points	500 words, rows scaled to length 1 (Day 86 compared directions).
t-SNE	perplexity 30, 1,000 steps, learning rate 100, early exaggeration 12 for 250 steps, five random starts.

Why this test

A clustered-looking picture proves nothing by itself. The same two numbers, measured in 50-D, say how much of the original structure each picture kept.

ACTUAL OUTPUT

```
space                    group check  preserved
50-D (no projection)           0.967      1.000
PCA, 2-D                       0.433      0.151
t-SNE, 2-D, seed 0             0.944      0.510
t-SNE, 2-D, seed 1             0.956      0.502
t-SNE, 2-D, seed 2             0.956      0.503
t-SNE, 2-D, seed 3             0.956      0.490
t-SNE, 2-D, seed 4             0.956      0.503
t-SNE's final KL per seed: 0.972, 0.984, 0.960, 1.005, 0.974
Actual output from this lesson's run.
```



Left: PCA's spread per direction. Then PCA, t-SNE (seed 0) and t-SNE of the untrained table, the same 500 words; grey are the 410 other words.

What it means

PCA keeps 0.433 on the group check and only 0.151 of near neighbours: the groups smear into each other. t-SNE keeps the groups (0.944–0.956) and about half of the near neighbours (0.490–0.510), and its five random starts give different layouts but the same numbers.

WHY CHOOSE THIS?

t-SNE shows neighbourhoods faithfully, and the numbers prove it.

WHY NOT THIS? (TRADEOFFS)

Half of each word's neighbours still change; no 2-D picture keeps them all.

4. Controls: what the pictures cannot be trusted with

DEFINITION — start here, no prior knowledge assumed

(a) t-SNE of Day 86's untrained random table, the same 500 words. (b) Three perplexities. (c) Made-up clusters in 50-D with known sizes and distances: A ten times as wide as B, and C four times as far from A as B is.

REAL-WORLD SCENARIO

Testing a thermometer in ice water and boiling water before trusting its readings.

ACTUAL OUTPUT

```
(a) The same 500 words with Day 86's untrained table (seed 0's random start, rows scaled to length
    1).
random table: group check 50-D 0.156, t-SNE 2-D 0.167; preserved 0.278
(b) Perplexity, on the trained table (seed 0):
  perplexity   5: group check 0.956, preserved 0.463
  perplexity  30: group check 0.944, preserved 0.510
  perplexity 100: group check 0.956, preserved 0.497
(c) Made-up clusters in 50-D whose true sizes and distances are known: A and B, 100 points each,
    A's spread 10 times B's; C, 100 points, 4 times as far from A as B is.
  50-D (true)  size A / size B =  10.09   distance A-C / distance A-B =   4.00
  PCA, 2-D     size A / size B =  12.79   distance A-C / distance A-B =   4.01
  t-SNE, 2-D   size A / size B =   0.99   distance A-C / distance A-B =   0.98
Actual output from this lesson's run.
```

What it means

(a) On random vectors the group check is chance (0.156 in 50-D, 0.167 after t-SNE), and the right-hand panel shows no groups: t-SNE did not invent any here. (b) Perplexity 5 to 100 barely matters here (0.463 to 0.510 preserved). (c) t-SNE gets the clusters right but draws A and B the same size (0.99 instead of 10.09) and C as close as B (0.98 instead of 4.00). PCA keeps both, roughly (12.79 and 4.01).

WHY CHOOSE THIS?

Read t-SNE for 'who is near whom'.

WHY NOT THIS? (TRADEOFFS)

Never read cluster sizes or the gaps between clusters off a t-SNE picture.

Interview preparation — technical questions

Q: How do you compute PCA?

A: Centre the data; eigenvectors of the covariance matrix (or the SVD of the centred data); keep the largest.

Q: What does perplexity control in t-SNE?

A: The width of each point's Gaussian: roughly how many neighbours it pays attention to.

Q: Why does t-SNE use a Student-t in 2-D?

A: Its heavy tails let moderately distant points sit far apart, which avoids crowding.

Q: How would you check a projection?

A: Measure neighbourhood preservation or a group check in both spaces, as here.

Q: Why run t-SNE several times?

A: Different starts give different layouts; only what repeats can be trusted.

Interview preparation — theoretical questions

Q: Why can't PCA show the word groups here?

A: Two linear directions keep under 40% of the spread; the groups differ along others.

Q: Can cluster sizes in a t-SNE plot be compared?

A: No: here a cluster ten times wider was drawn the same size.

Q: Does t-SNE invent clusters?

A: It can, with some settings and data; here, on the random table, it drew none and the group check stayed at chance.

Q: When is PCA the better picture?

A: When distances and spreads matter, or the data really lies near a plane.

Key takeaway and what's next

> **KEY TAKEAWAY**
> A picture of embeddings needs a number beside it. PCA keeps distances honest but, for these vectors, loses the groups; t-SNE keeps the groups and half of each word's neighbours, but in the known-cluster test it drew a cluster ten times wider at the same size, and a gap four times wider at the same width.

- Group check: 50-D 0.967, PCA 0.433, t-SNE 0.944-0.956 (five starts), random table 0.167.
- Neighbours preserved (10 nearest): PCA 0.151, t-SNE about 0.50.
- Known clusters: true size ratio 10.09, t-SNE 0.99; true distance ratio 4.00, t-SNE 0.98.

Suggested watch order, if starting from zero

- 1. Principal Component Analysis (PCA), Step-by-Step (StatQuest) — PCA, step by step.
- 2. t-SNE, Clearly Explained (StatQuest) — t-SNE's neighbour probabilities and Student-t curve.

Day 88 preview

- **Words never seen.** fastText builds a word's vector from its character n-grams, so a word missing from training still gets one. Tested on words removed from the corpus on purpose.